DANIEL ADEWOLE ADEDIRAN

Genomics & Bioinformatics | One Health | Antimicrobial Resistance | Training & Capacity Building

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PROFESSIONAL PROFILE

Public health and bioinformatics professional (MPH; B. Tech Biochemistry) working where genomics meets public health in Nigeria. I teach and apply command-line AMR gene detection, multiple sequence alignment and phylogenetic, SARS-CoV-2 genomic epidemiology with Nextclade and Nextstrain, differential gene expression analysis, and structural bioinformatics, and I have analyzed antimicrobial use data across six states. My research background includes computational vaccine and diagnostic design for infectious and zoonotic pathogens (Nipah virus, Lassa virus, Treponema), dog metagenomics metadata analysis, and high-performance computing.

EDUCATION

Master of Public Health (MPH) | Bingham University *Jan 2025–Nov 2025*

Coursework: Epidemiology, Biostatistics, Research Methodology, Health Informatics, Implementation Science, Global Health, Health Policy, Monitoring & Evaluation, Project Management.

B. Tech, Biochemistry | Ladoke Akintola University of Technology (LAUTECH) *2018 – 2023*

- CGPA: 4.36/5.0. Relevant coursework: Molecular Genetics, Bioinformatics.
- Academic project: Assessment of Zingiber officinale on Tramadol-Induced Nephrotoxicity in Rats.

CORE COMPETENCIES

- **Genomics & sequence analysis:** AMR gene detection from FASTA sequences (abricate, argNorm, seqkit); multiple sequence alignment (MAFFT, Clustal Omega, MEGA); phylogenetic inference (IQ-TREE); reproducible command-line workflows.
- **Genomic epidemiology:** SARS-CoV-2 lineage assignment and phylogenetic interpretation with Nextclade and Nextstrain; public health outbreak analysis in R.
- **Transcriptomics & omics:** Differential gene expression analysis in R (DESeq2); Immunoinformatics and proteomics-based work (VaxiJen, IEDB); exposure to single-cell and spatial omics communities.
- **Structural bioinformatics:** PDB, AlphaFold structure prediction in Google Colab, I-TASSER, and web-based molecular docking and simulation.
- **Computing:** Linux/Bash, Git/GitHub, R, Python (pandas, matplotlib, scikit-learn), SQL, SPSS, SAS, HPC clusters via SSH, LaTeX/Overleaf, Quarto.
- **One Health & public health:** Antimicrobial use and resistance surveillance, climate-sensitive infectious diseases, epidemiology and biostatistics, WASH needs assessment.
- **Teaching & capacity building:** Practical session and curriculum design, cohort programme delivery, mentoring, learning management system content; 270+ learners and professionals trained.
- **Scientific communication:** Manuscripts, posters, conference and seminar presentations, technical reports and teaching documents.

PROFESSIONAL EXPERIENCE

Health Data and Bioinformatics Analyst | DataLab Analytics *Jan 2025 – Dec 2025*

Remote, Nigeria

- Designed and taught a two-part command-line AMR genomics module: detecting resistance genes in FASTA sequences with abricate against ResFinder, CARD and NCBI, then harmonising gene names to a shared ontology (ARO) with argNorm and building presence/absence matrices and heatmaps for interpretation in a surveillance setting.
- Taught multiple sequence alignment and phylogenetic inference with MAFFT and IQ-TREE, with learner projects on tree visualisation and interpretation.
- Introduced genomic epidemiology through SARS-CoV-2 case studies using Nextclade and the Nextstrain web platform.

- Taught structural bioinformatics, covering the PDB, AlphaFold structure prediction in Google Colab, I-TASSER, and web-based molecular docking and simulation tools.
- Taught Linux and command-line fundamentals, differential gene expression analysis in R, and public health outbreak analysis in R; prepared a DESeq2 RNA-seq teaching document and marked a bioinformatics assignment on the pasilla dataset.
- Conceived and led SHE Innovate, an end-to-end data skills cohort programme for 50 women, covering recruitment, curriculum design, delivery and mentorship; 96% completed the programme and 90% reported immediate application of the skillset.
- Trained and mentored 70+ learners; built curriculum modules and learning management system content (videos, documentation, structured learning paths); reviewed and optimised R, Python and Linux scripts used in training.

Consultant | Ajisefini Consulting *Dec 2025 – Jul 2026*

Remote, Nigeria

- Analysed antimicrobial use data in Stata across six states as part of antimicrobial use and resistance surveillance work.
- Analysed ACE3 HIV service and eMTCT data and contributed to a knowledge product on service continuity under USAID funding changes.
- Designed data collection instruments, then cleaned and analysed data in R and Python (regression, hypothesis testing), producing client-ready reports with Git-versioned, reproducible workflows.

R&D Assistant II and Teaching Assistant | Helix Biogen Institute *Aug 2021 – Present*

Ogbomoso, Nigeria

- Contributed to peptide-based diagnostic kit development for Treponema spp. using in silico bioinformatics and immunoinformatics approaches.
- Led 17 research interns on computational vaccine design for Nipah virus and COVID-19, coordinating literature reviews, protocol adherence and analytical workflows that fed into 2 grant proposals.
- Began as a Research Intern (Aug – Dec 2021), applying reverse vaccinology and structural bioinformatics to diagnostic and vaccine design for infectious diseases; carried out phylogenetic and multiple sequence alignment analysis with MEGA and Clustal Omega.
- Developed standard operating procedures for vaccine design pipelines, documented and validated workflows, and ran weekly review sessions to keep projects on schedule.
- Trained 200+ participants on MEGA, VaxiJen and IEDB for vaccine design and immunology research, and prepared hands-on materials and exercises.

RESEARCH PROGRAMMES & FELLOWSHIPS

Co-Lead | ASBCB Omics Codeathon *Aug 2022 – Nov 2022*

- Led a 5-person team to identify differentially expressed genes in Multiple Sclerosis datasets using R.
- Coordinated weekly code sprints and documented the analysis on GitHub for reproducibility.
- Presented results to NIH-affiliated mentors and drafted a manuscript outline for further publication.

Scholar | AI-BOND Program *Dec 2023 – Apr 2024*

- Used Texas Advanced Computing Center clusters (Lonestar and Rancher) via SSH for computational analysis.
- Worked in Python, R and SAS on cross-team statistical modelling and GWAS studies; co-led an analysis of Framingham Heart Study data on cardiovascular risk predictors.
- Delivered weekly presentations on neurodegenerative mechanisms.

Intern | BigDataBiology, Fudan University (Remote) *May 2023 – Oct 2023*

- Analysed and visualised genomic metadata from Shanghai dog samples in Python (pandas, matplotlib) for the dog metagenomics team.
- Prepared internal reports and visual dashboards that supported project discussions; used Git for version control and collaborative code sharing.
- Refined analysis methods and presentation of findings in weekly feedback meetings with mentors.

Mentee | KaggleX BIPOC Mentorship Program (Remote) *Jul 2023 – Oct 2023*

- Learned scikit-learn and TorchDrug for bioinformatics-focused machine learning and contributed to a CRISPR repair outcome prediction model.
- Applied transfer learning approaches for drug discovery applications and presented updates in bi-weekly mentor check-ins.

Member, Data Federation Training Group | AgBioData *Jan 2023 – Dec 2023*

- Compiled technical documentation on iRODS-based data federation workflows and helped standardise training resources across four working groups.
- Facilitated a pilot session to test the materials and mentored new contributors to the training team.

TRAINING, COMMUNITY & PROGRAMME EXPERIENCE

Training & Education Committee Member | CSIDNet *Jun 2025 – Present*

- Help the Climate Sensitive Infectious Diseases Network design and coordinate capacity-building programmes on climate-sensitive infectious diseases, including training objectives, content and delivery formats.
- Take part in planning, monitoring and evaluating training activities so they stay aligned with the network's knowledge-sharing goals.
- Support preparation of educational and communication materials that explain complex climate and infectious disease concepts to different audiences.

Teaching Assistant | H3ABioNet *Jul 2022 – Oct 2022*

- Introduced participants to bioinformatics, Linux environments and biological databases in an introductory training course; gave feedback on assignments and quizzes.

Scientific Manager I | Panoramics-Vision *Jun 2023 – Jan 2024*

- Built and maintained a database of African single-cell omics researchers to support networking and collaboration.
- Curated and shared 20+ peer-reviewed articles and coordinated a monthly spatial transcriptomics series; tracked engagement to refine content.

Needs Assessment Intern, MERLA Branch | Global Hart *Feb 2026 – May 2026*

- Supported needs assessment work for the Haiti WASH Education Program and produced five formal deliverables, including a storytelling-as-methodology presentation.
- Developed a sector-agnostic Needs Assessment Toolkit field guide with in-text citations and global resource references.

Program Officer | SDGs CDS, NYSC – Tafa LGA *Dec 2023 – Nov 2024*

- Designed and ran local awareness programmes and webinars on climate action and sustainability, working with local government and community groups.
- Drafted concept notes, programme summaries and reports; shared 10+ funding and training opportunities with fellow corps members.

TEACHING MATERIALS DEVELOPED

- AMR Gene Detection from FASTA Sequences, a two-day LaTeX/Overleaf tutorial set in a Northwest Nigerian clinical scenario: Day 1 covers sequence acquisition, quality control and screening with abricate and seqkit; Day 2 covers argNorm harmonisation, a combined hit table, a presence/absence matrix and heatmap, and guided interpretation of what the results do and do not show.
- Multiple sequence alignment and phylogenetic practical (MAFFT, IQ-TREE) with an assigned visualisation project for learners.
- Introduction to Nextstrain and Nextclade using SARS-CoV-2 genomic case studies.
- Structural bioinformatics practical covering the PDB, AlphaFold in Google Colab, I-TASSER, and web-based docking and simulation; teaching documents for differential gene expression (DESeq2) and outbreak analysis in R.

PUBLICATIONS & SCIENTIFIC PRESENTATIONS

- Co-author, Frontiers in Oncology (2024): epigenetic modifications in solid tumor metastasis in people of African ancestry.
- Co-author on publications in Genes & Genomics, Vaccines and Immunogenetics covering proteomics, mRNA vaccine design, RNA-seq and genomic surveillance; contributor to work on COVID-19 vaccine technology transfer and proteome-based analysis of SARS-CoV-2 variants.
- Manuscripts in preparation: ACE3 eMTCT programme performance across three Northwest Nigeria states, and an evaluation of NFELTP across 10 cohorts (2008–2018).
- Presented a computational mRNA vaccine design for Lassa virus at the Beyond Science Initiative Conference (2023); posters on rotavirus genetic diversity in Africa and diagnostic kit design for Schistosoma haematobium.
- Full publication list available on request or via Google Scholar.

CERTIFICATIONS & SHORT COURSES

- Climate and Health Data Science (Data.org)
- Train the Trainer Workshop (H3ABioNet, Wellcome Connecting Science & EMBL-EBI)
- Introduction to Epidemic Theory (Epiverse); Introduction to Global Burden of Disease (IHME); Time Series Analysis
- Implementation Science Workshop (Bingham University); Implementation Science in Public Health (ISGlobal)
- Monitoring and Evaluation Essentials (Disaster Ready); Project Management Professional (ICBMD)

GRANTS, AWARDS & SCHOLARSHIPS

- Fair Forward Best Methodological Rigor Award (2025)
- Poster Award, Nnamdi Azikiwe Bioinformatics Genomics Consortium (2022); RNA Therapeutics Symposium Scholarship (2022)
- Dan Kohn Scholarship, KubeCon + CloudNativeCon Europe (2023); KaggleX Career Development Scholarship (2023)
- eDNA Micro-grant for R Package Development Course, Physalia (2023); Africa Rotavirus Conference Scholarship (2023)

PROFESSIONAL AFFILIATIONS

H3ABioNet Trainer Network, Carpentries Instructor Community, African Society for Bioinformatics and Computational Biology (ASBCB), International Society for Computational Biology, Human Cell Atlas Community, Climate Sensitive Infectious Diseases Network (CSIDNet), AMMnet Nigeria.

REFERENCES

Available upon request.